Table 1.2 Calculation of number of entities in one mole of substance.

S. No.	Name of substance	Mass of the substance taken (gram)	Mass of single atom or molecule (gram) = Atomic mass or Molar mass / Avovadro Number	No. of atoms or molecules = Mass of substance ÷ Mass of single atom or molecule
	(1)	(2)	(3)	(2)÷(3)
1.	Elemental Carbon (C-12)	12	1.9926×10^{-23}	$\frac{12}{1.9926 \times 10^{-23}} = 6.022 \times 10^{23}$
2.	Glucose ($C_6H_{12}O_6$)	180	29.89×10^{-23}	$\frac{180}{29.89 \times 10^{-23}} = 6.022 \times 10^{23}$
3.	Potassium dichromate ($K_2Cr_2O_7$)	294.18	48.851×10^{-23}	$\frac{294.18}{48.851 \times 10^{-23}} = 6.022 \times 10^{23}$
4.	Potassium permanganate ($KMnO_4$)	158.03	26.242×10^{-23}	$\frac{158.03}{26.242 \times 10^{-23}} = 6.022 \times 10^{23}$

From the calculations we come to know that 12 g of carbon-12 contains 6.022×10^{23} carbon atoms and same numbers of molecules are present in 158.03 g of potassium permanganate and 294.18 g of potassium dichromate. Similar to the way we use the term 'dozen' to represent 12 entities, we can use the term 'mole' to represent 6.022×10^{23} entities (atoms or molecules or ions)

One mole is the amount of substance of a system, which contains as many elementary particles as there are atoms in 12 g of carbon-12 isotope. The elementary particles can be molecules, atoms, ions, electrons or any other specified particles.



Gastric acid and antacids:

Antacids are commonly used medicines for treating heartburn and acidity. Do you know the chemistry behind it?

Gastric acid is a digestive fluid formed in the stomach and it contains hydrochloric acid. The typical concentration of the acid in gastric acid is 0.082 M. When the concentration exceeds 0.1 M it causes the heartburn and acidity.

Antacids used to treat acidity contain mostly magnesium hydroxide or aluminium hydroxide that neutralises the excess acid. The chemical reactions are as follows.

$$3\ HCl + Al(OH)_3 \rightarrow AlCl_3 + 3\ H_2O$$

$$2\ HCl + Mg(OH)_2 \rightarrow MgCl_2 + 2\ H_2O$$

From the above reactions we know that 1 mole of aluminium hydroxide neutralises 3 moles of HCl while 1 mole of magnesium hydroxide neutralises 2 moles of HCl.

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